

Nonvalidation of mesospheric horizontal wind fluctuations derived from temperature data via comparison with MF radar wind measurements

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Abstract

Mesospheric transport processes and energy budget are tied closely to gravity wave activity. Any means of monitoring dynamics reliably can only contribute to our understanding of the physics of the mesosphere. In this study we employ temperature data derived from hydroxyl (OH) emission from around 85 km and combine it with the theory of *Hines* [1960] to investigate the practicality of deducing fluctuations in horizontal wind speed. *Myrabø and Harang* [1988] have used a simplified theory [*Hines*, 1965] but, apparently, to deduce tidal fluctuations in the horizontal wind; this prompts us to investigate the range of timescales for which the *Hines* [1965] result and the assumptions implicit therein may be used in practice. Our conclusion is that this theory may not be extended to scales of dynamics outside those intended by *Hines*. Thus while the *Myrabø and Harang* [1988] results seem plausible, they are probably incorrect. © 1999 American Geophysical Union

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